Aansh Singh

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EDUCATION

University of California, Los Angeles | Henry Samueli School of Engineering **Los Angeles, CA**
Bachelor of Science in Computer Science *Expected Graduation: June 2028*

EXPERIENCE

Software Engineer **2025 – Present**
Bruin Underwater Robotics (BUR) @ UCLA *Los Angeles, CA*

- Developed Python/YOLOv11 object-detection models for autonomous RoboSub perception, recognizing path markers, slalom gates, maps, and obstacles.
- Automated a Unity simulation pipeline to generate 3,000+ labeled images and bounding boxes for underwater perception model training.
- Exported PyTorch models to ONNX for TensorRT acceleration on NVIDIA Jetson; integrated Jetson and Raspberry Pi via ROS2 for onboard perception and real-time motor control.
- Collaborated with software, electrical, and mechanical engineers to integrate perception, computing, sensors, and propulsion.

Software Engineering Team Lead Intern **May 2024 – Sept 2025**
D-Tech *Remote*

- Led 5 interns designing Flippper, an ePaper education platform; translated a German patent into software requirements, user flows, component interactions, and functional specifications.
- Defined ePaper workflows, interface behavior, and hardware/software integration points with the founder and an Oracle mentor to assess technical feasibility.
- Directed UI/UX prototypes, 3D models, and visualizations to validate workflows and explain system behavior.

Instructor **May 2024 – Jan 2025**
CompuChild *Austin, TX*

- Taught Python and Scratch to middle school students through hands-on coding exercises and project-based lessons.
- Guided line-following robot and solar-car projects; debugged student code and devices.

PROJECTS

CodeGraph: Repository Intelligence | *TypeScript, React, Python, FastAPI, Neo4j, LangGraph*

- Built a full-stack React/TypeScript and Python/FastAPI repository intelligence app with GitHub ingestion, 32-way bounded Tree-sitter parsing, Neo4j file/function/call graphs, and NDJSON progress streaming.
- Developed a Claude-powered LangGraph ReAct agent with 7 repository-scoped tools for dependency, blast-radius, external-call, architecture, source, and vector-search analysis.
- Built Graph-RAG with Neo4j vector indexes, 1,536-dimensional OpenAI embeddings, and GDS Leiden subgraphs; reduced LLM context 97.5% from 737K to ~18K tokens per query.
- Automated Postman API tests for ingestion, graph retrieval, Graph-RAG, and cleanup with NDJSON assertions, negative cases, and HMAC-SHA256 webhook validation; verified with 82 Pytest cases and Node tests.

TaskForge: Fault-Tolerant Task Processing | *Go, PostgreSQL, Redis, Docker, Prometheus*

- Engineered Go/PostgreSQL task processing with atomic SQL claims, priority scheduling, heartbeats, renewable leases, backoff retries, and idempotent submission; added Prometheus/Grafana observability.
- Tested task-state integrity and crash recovery across 3,000 tasks / 6,000 ordered attempts with zero duplicates; recovered 30 hard-killed attempts with zero duplicate effects.
- Benchmarked 15.84× scaling across 16 workers at 99% efficiency with Prometheus reconciliation; persisted 30,000/30,000 API submissions at a 2,143/sec median with zero request/transport errors.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Java, SQL, C++, C, HTML/CSS
Web / APIs: React, React Flow, FastAPI, REST APIs, Postman, Pytest, Node Tests, Git
Backend / Data: PostgreSQL, Redis, Docker, Prometheus, Grafana, Neo4j GDS, Tree-sitter
AI / Robotics: LangGraph, RAG, OpenAI Embeddings, PyTorch, YOLOv11, ONNX, TensorRT, ROS2, NVIDIA Jetson, Raspberry Pi, Unity